EARTHPRESS · FEASIBILITY ASSESSMENT

EarthPress feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First feasibility assessment (kit 1.8 pipeline) for option A after the kit 1.9.1 upgrade

EarthPress feasibility assessment

Feasible on paper: the lever and toggle press is calculated, drawn at detail level D3, priced and planned for a small welding shop, and its one structural shortfall (the toggle pins) is fixed; what remains uncertain is the soil, not the machine, and the press is 3.8 kg over its mass target and USD 42 over its cost target. Method: STANDARDS section 23.

Table 1. Feasibility by aspect (option A, the current design).

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	The mechanism and every load-path member are calculated at the 173 kN design load (lowest factor 1.29); kinematics, grip heights and eject checked; the detailed model passes the D3 fidelity check with no interference	High	The soil compaction law is assumed (STM S6), so the peak pull and the fill window could shift; two operators are needed	EPR-STM-001, EPR-CAL-001, `cad/fidelity.json`
Materials	Mild steel plate, UPN 80, tube and pipe from a stockist; 42CrMo4 +QT pins and case-hardened bushes bought turned	High	42CrMo4 bar and 40 x 46 mm bushes to confirm locally (EPR-DEC-001, to confirm item 2)	BOM lines 1 to 7
Manufacturing and assembly	Sawing, cutting, drilling, tapping, stick welding and grinding; 14 components, 13 assembly steps with a picture each; no machining in the shop	High	Squareness of the mold cavity (ground within 0.5 mm) and the 420 mm connecting link centres need care	EPR-BLD-001, DWG-101 to DWG-115
Power and controls	Human powered; no controls	High	Two people on the T-handle at up to 405 N each with the softer design soil	PR-005, PR-015
Supply of parts	Common sections and fasteners; turned pins from a machine shop	Medium	Prices are indicative; the BOM is USD 542 against USD 500	`bom/bom.csv`, EPR-CAL-001 section 11
Mass and transport	203.8 kg in pieces of 44.5 kg or less; fits a 1.5 m pickup bed without the lever	Medium	Over the 200 kg R7 total (decision D2)	EPR-CAL-001 section 8
Blocks	2 MPa is the low end of CSEB practice; strength depends on soil and curing	Medium	Block strength and wear can only be verified at TRL 4 (on hold)	STM S2, S18; PR-003, PR-008, PR-010

Ratings: High, Medium, Low.